

Course Syllabus**GRAD-E1282: Natural Language Processing****1. General information**

Course Format	Onsite
Instructors	Luis Fernando Ramírez Ruiz , M.Sc.; TA: TBA
Instructor's E-mail	l.ruiz@hertie-school.org
Instructor's Office Hours	Onsite by appointment.

Link to [Study, Examination and Admission Rules and MIA, MDS and MPP Module Handbooks](#)

For information on **course room, times and session dates**, please consult the [Course Plan](#) on *MyStudies*.

Instructor Information:

Luis Fernando is a doctoral candidate and researcher at the Robert Koch Institute. His line of research focuses on using machine learning method, unstructured and unlabeled multimodal data to understand and model narratives on Climate Change and Public Health at scale.

At the Hertie School, he was a Research Associate as a part of the blame attribution project leveraging LLMs to analyze political discourses and worked as a Teaching Assistant for the Natural Language Processing course. He earned an MPP and MDS at the Hertie School. In Mexico, he studied Political Science (ITAM) and Philosophy (UNAM).

2. Course Contents and Learning ObjectivesCourse contents:

The course provides a conceptual and applied introduction to Natural Language processing (NLP). The program introduces NLP within the broader fields of Machine Learning and Deep Learning. It will cover core topics such as text representation, text preprocessing, supervised and unsupervised learning with labeled and unlabeled datasets, model evaluations, and emerging trends including using other data sources such as audio and video.

By the end of the course, students will understand the main ideas behind NLP and will be able to apply these tools to their own research questions or applied projects. The lecture will include applied coding sessions using Python as a programming language and VSCode as the recommended development environment. As an advanced seminar, the course requires prior knowledge of Causal Inference or Statistics II, and programming experience in Python or R at the level of "Introduction to Data Science".

Main learning objectives:

At the end of this course, students should:

- (1) Have a clear understanding of NLP concepts, algorithmic tools, and assumptions to appropriately build NLP systems for research or other objectives.
- (2) Be able to specify and implement NLP applications using Python.
- (3) Know how to interpret, evaluate, and communicate NLP applications.
- (4) Be able to explore and develop more complicated/realistic NLP systems independently or in the context of further seminars.

Software:

To work through this course, a working installation Python is required. In class (when required) and during lab sessions, students will work with the integrated development environment Visual Studio Code (VSCode) or other IDEs (PyCharm). All software is available online for free.

Target group:

Aimed at students who have a keen interest in learning new programming languages, applying tools from statistical inference within the realm of machine learning and deep learning.

Teaching style:

This is a methods course that combines lecture-style teaching with the practical and joint implementation of code in Python during separate lab session. The lecture is taught by Luis Fernando Ramírez. The lab sessions will be led by the TA (TBA). Students are encouraged to ask questions at any time and are also confronted with questions by the lecturer during the course.

Prerequisites:

The course assumes prior knowledge of statistics at the level of "Statistics II: Statistical Modeling and Causal Inference", including basic probability principles, familiarity with standard notation for statistical inference such as (generalized) linear regression, and basic matrix operations. Additionally, prior knowledge of R or Python programming languages at the level of "Introduction to Data Science" is expected. Knowledge of control structures, i.e., if-else and loops, as well as data types and structures is essential and recommended.

3. Grading and Assignments

Policy on the use of AI tools for assignments

The use of generative AI tools (such as ChatGPT) is **"permitted throughout this course"**. Students must adhere to the specific instructions provided for each assignment. Where AI tools are used, students are expected to clearly indicate how they were used and to ensure the originality and academic integrity of their work. **Students must include a Statement of Authorship in all written assignments.** Improper or undisclosed use of AI tools may be considered a violation of academic integrity policies. Please consult the course instructor in case of doubt. Further guidance is available in the [Artificial Intelligence Tools at the Hertie School - Teaching Guidelines for Instructors and Students](#), including a template for the **Statement of Authorship**, under section 1.1.

Composition of Final Grade:

Assignment	Due	Submission	Grade
Assignment 1: Class Participation	-	Moodle	10%
Assignment 2: Quizzes (Q)	Deadline: Wednesday in the week after the seminar until 11:59pm	Moodle	3 x 10%
Assignment 3: Outline for Research Note	Deadline: One week after session 7	Submit via Moodle	20%
Assignment 4: Research Note	Deadline: 23 December until 11:59pm	Submit via Moodle	40%

Assignment Details:

Assignment 1: Class participation

Content discussed in class requires a prior reading of the assigned required literature to enrich the dialogue between students and lecturer. Participation is marked by being active, consistent, and meaningfully contributing to the discussion on concepts and ideas covered in class.

Assignment 2: Quizzes

Starting from session 3 and until session 10, there will be an online quiz every two sessions, contributing 10% each to the final grade. Quizzes will assess the acquired knowledge and comprehension related to the material of the last two sessions. A "Q" in the session overview below indicates quiz dates.

Assignment 3: Outline for Research note

The purpose of the assignment, contributing 20% to the final grade, is to promote students to actively plan the research note. It is expected to include a research question, data (including data recollection strategy), define the natural language processing pipeline to start the analysis. If applicable, they can also define causal mechanisms when their research question is part of a causal inference research design.

Assignment 4: Research note

In the research note, contributing 40% to the final grade, students will apply natural language processing in practice using Python to pursue their own research question with data of their choosing using the steps and tools covered throughout this course. Students are free to choose one of the two scenarios for this assignment. Findings must be reported in a research note (approx. 10 pages, excluding the title page and references, and no more than 2 figures in main text). Students are strongly encouraged to discuss their plans for the research note with the instructor during the office hour.

Late submission of assignments:

For each day the assignment is turned in late, the grade will be reduced by 10% (e.g. submission two days after the deadline would result in 20% grade deduction).

Attendance:

Students are expected to be present and prepared for each class session. Active participation during lectures and seminar discussions is essential and recommended. Please note that students can miss up to two sessions (out of twelve) if no course assignments are affected. **Students do not need to inform the course instructor(s) in advance if they are unable to attend a session.** For further information please consult the [Examination Rules](#) §10. Academic Integrity:

The Hertie School is committed to the standards of good academic and ethical conduct. Any violation of these standards shall be subject to disciplinary action. Plagiarism, misuse of AI, free riding in group work, and other deceitful actions are not tolerated. See [Examination Rules](#) §16, the Hertie [Plagiarism Policy](#), and [the Hertie Guidelines for Artificial Intelligence Tools](#).

Compensation for Disadvantages:

If a student furnishes evidence of being unable to take an examination as required in whole or in part due to disability or permanent illness, the Examination Committee may upon written request approve learning accommodation(s). In this respect, the submission of adequate certificates may be required. See [Examination Rules](#) §14.

Extenuating circumstances:

An extension can be granted due to extenuating circumstances (i.e., for reasons like illness, personal loss or hardship, or caring duties). In such cases, please contact the course instructor and Examination Office *in advance* of the assignment deadline.

4. **Session Overview**

Session	Session Title
1	Introduction, motivation and course overview
2	String processing, regular expressions, and text preprocessing
3	Text representation I - Vector-based (Q)
4	Text representation II - Distribution-based
5	Basic models I - Text classification with logistic regression
6	Neural models I - Text classification with neural networks (Q)
7	Neural models II - Topic modelling with LDA + BERTopic
8	Neural models III - Language modelling with Transformers I
9	Neural models III - Language modelling with Transformers II (Q)
10	NLP visualizations + Explainable AI
11	Extending NLP to LLM workflows and agents
12	LLM Workflows and Agents II

5. **Course Sessions and Readings**

All mandatory course readings can be accessed on the course Moodle page.

Session 1: Motivation and course overview

During the first meeting, we will explore the definition of and rationale behind natural language processing and discuss its potential applications. Toward the end of the session, we will look ahead and clarify organizational matters.

Required Reading	- Vajjala et al. (2020). Chapter 1
Optional Reading	- Hobson/Dyshel (2024). Chapter 1, 1-1.3.1

Session 2: String processing, regular expressions, and preprocessing

This session is fully dedicated to regular expressions, the fundamental tool for describing text patterns. Regular expressions offer an algebraic notation for characterizing a set of strings and lie at the heart of text normalization and many algorithmic tools in NLP.

Required Readings	- Jurafsky/Martin (2024). Section 2.1 and Hobson/Dyshel (2024). Chapter 2, 2-2.9
Optional Readings	- Hobson/Dyshel (2024). Chapter 1, 1.4-1.6.1 and Jurafsky/Martin (2024). Sections 2.2-2.7
Core Python Tools	re, nltk, sentencepiece, scikit-learn, spacy, tokenizers

Session 3: Text representation I - Vector-based

In our first session on text representation, we begin to develop an intuition for representing text numerically. To this end, we consider common vectorization approaches, such as One-Hot-Encoding, Bag Of Words, and TF-IDF. Following a conceptual introduction, we implement these text representation approaches using core Python libraries for NLP.

Required Readings	- Vajjala et al. (2020). Chapter 3, “Basic Vectorization Approaches”
Optional Readings	- Hobson/Dyshel (2024). Chapter 3
Core Python Tools	numpy, pandas, scikit-learn

Session 4: Text representation I - Distribution-based

We continue our inquiry into text representations and build on the previous session by focusing on distribution-based representations. In particular, we consider low-dimensional learned text representations called embeddings, their architectural variants, and how to implement them in Python.

Required Readings	- Vajjala et al. (2020). Chapter 3, “Distributed Representations” and “Distributed Representations Beyond Words and Characters”
Optional Readings	- Hobson/Dyshel (2024). Chapter 6

Core Python Tools	fasttext, gensim
--------------------------	------------------

Session 5: Basic models I - Text classification with logistic regression

This session introduces basic yet highly useful and widely applied supervised models for natural language processing. We discuss different classification approaches, such as logistic regression and support vector machines, and demonstrate the practical setup and training routine in the context of logistic regression applied to text classification tasks.

Required Readings	- Vajjala et al. (2020). Chapter 4, up to and including "Support Vector Machine"
Optional Readings	- Raschka et al. (2022). Chapter 3, "Modeling class probabilities via logistic regression"
Core Python Tools	imblearn, scikit-learn

Session 6: Basic models II - Topic modelling with LDA + BERTopic

The second session on basic models for NLP introduces unsupervised models. Following a brief conceptual introduction of different algorithms, we focus on Latent Dirichlet allocation and its practical implementation to discover topics or meaning in texts.

Required Readings	- Hobson/Dyshel (2024). Chapter 4.
Optional Readings	- Raschka et al. (2022). Chapter 8, "Topic modelling with latent Dirichlet allocation" - Grootendorst (2022), "BERTopic: Neural topic modeling with a class-based TF-IDF procedure"
Core Python Tools	numpy, pandas, scikit-learn

Session 7: Neural models I - Text classification with neural networks

In this session we begin delving into more complex neural models. We start with a brief conceptual introduction of neural networks in general and 1-dimensional convolutional neural networks for text in particular. Finally, we introduce PyTorch by implementing the latter for the purpose of text classification.

Required Readings	- Raschka et al. (2022). Chapter 12, "First Steps with PyTorch", "Building input pipelines in PyTorch", and "Building an NN model in PyTorch"; Chapter 13, "Simplifying implementations of common architectures via the torch.nn module"; Chapter 14, "The building blocks of CNNs"
Optional Readings	- Raschka et al. (2022). Chapter 2 and 11
Core Python Tools	torch

Session 8: Neural models II - Language modelling with Transformers I

Further increasing the complexity of neural models for NLP, we begin discussing the anatomy of transformers with a particular focus on their precursors - recurrent neural networks for sequence modelling. On the practical side, we extend our PyTorch-based implementation of a neural network model for text classification from the previous section by adding corresponding RNN and LSTM layers.

Required Readings	- Raschka et al. (2022). Chapter 15 - Turnstall et al. (2022). Chapter 1, "The Encoder-Decoder Framework", "Attention Mechanisms"
--------------------------	---

Optional Readings	Tunstall et al. (2022). Chapter 3
Core Python Tools	torch

Session 9: Neural models II - Language modelling with Transformers II

In our final session on neural models, we delve deeper into transformers and introduce the Hugging Face ecosystem. Using the corresponding transformers library, we learn how to easily apply and fine-tune, i.e. apply transfer learning to, pre-trained large language models for the purpose of text classification.

Required Readings	- Tunstall et al. (2022). Chapter 1, "Transfer Learning in NLP", "Hugging Face Transformers", and "The Hugging Face Ecosystem"
Optional Readings	Tunstall et al. (2022). Chapter 2
Core Python Tools	torch, transformers

Session 10: NLP visualizations

In this session, we turn to visualization techniques for natural language processing. These range from confusion matrices, over word-clouds and 2-dimensional embedding visualizations to local interpretable model-agnostic explanations that allow for exploring the contribution of different textual components to a classification decision.

Required Readings	-
Optional Readings	-
Core Python Tools	captum, matplotlib, scikit-learn, tensorboard, wordcloud

Session 11: Extending NLP to LLM workflows and agents

The final couple of sessions explores how modern NLP techniques can be extended into broader AI applications by orchestrating large language models in workflows and agents. Core concepts of LangChain and LangGraph are introduced and illustrated through minimal, hands-on examples.

Core Python Tools	captum, matplotlib, scikit-learn, tensorboard, wordcloud
--------------------------	--

Session 12: Extending NLP to LLM workflows and agents

Required Readings	- TBA
--------------------------	-------

Optional Readings	- TBA
--------------------------	-------

Core Python Tools	langchain, langgraph
--------------------------	----------------------

Final Exam Week: no class